

MANNAN SHUKLA

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TECHNICAL SKILLS

Kubernetes (operators, controllers, CRDs), Docker, Helm, ArgoCD, Terraform, Cilium • Go, Python, Bash, JavaScript, TypeScript • vLLM, SGLang, TensorRT-LLM, Triton Inference Server, continuous batching, PagedAttention/KV cache, FP8/INT8 quantization, speculative decoding • CUDA, Slurm, DCGM, nvidia-smi, NCCL • PyTorch, Hugging Face, LoRA, LangChain • AWS (EC2, ECS, RDS, S3, DynamoDB, CloudWatch, IAM, SNS) • Prometheus, Grafana • GitOps, CI/CD, GitHub Actions • Auth0, JWT, OAuth, RBAC

EXPERIENCE

Software Engineer

July 2025 - Present

Voltage Park / Lightning AI

Redmond, WA

- Built **VP Cloud** platform features in **Go**, including **Kubernetes** controllers for fleet/node lifecycle backed by an event-sourced **DynamoDB** model (composite keys, org-scoped GSIs)
- Designed hot weight reloading for **BYOM vLLM** deployments: a **Go** sidecar calling vLLM's `reload_weights/sleep-mode` APIs to swap fine-tuned checkpoints in-place in seconds, eliminating 10–30 min GPU redeployments
- Designed and shipped a **P2C** load balancer for the distributed storage control plane, improving request distribution across backend instances
- Built **GitOps** delivery (**ArgoCD**) and infrastructure-as-code (**Terraform**, **Helm**) to provision multi-tenant **Kubernetes** clusters, with CI/CD via **GitHub Actions**
- Maintain a multi-tenant managed **Kubernetes** platform, reviewing PRs across **Cilium/Envoy** networking, **NVIDIA GPU Operator** health checks, **kube-prometheus-stack** observability, and tenant-isolation hardening
- On-call for production **H100 Kubernetes** clusters; debugged container networking (**WireGuard**, **CoreDNS**, **Cilium**), **MPI Operator**, and **GPU Operator** issues
- Built **Prometheus/Grafana** fleet observability and a **Slurm** prolog/epilog GPU health-check suite (ECC, PCIe AER, NCCL stalls, thermal) with automatic node draining

Independent Researcher: SuperAgent

Apr 2024 - Apr 2025

Rutgers University

New Brunswick, NJ

- Built **LLM**-based agent recommendation system under Prof. Yongfeng Zhang; achieved 83% accuracy vs. **GPT-4** ground truth using a 1B model via **LoRA** fine-tuning with **Unsloth**

Partnerships + Business Development

Feb 2025 - May 2025

Nas.io

New York, NY

- Drove influencer partnerships and client onboarding for a Series A startup, generating \$2K GMV in one month while 50x'ing outreach efficiency via a **Python/Klaviyo** automation pipeline

PROJECTS

BYOA: Bring Your Own Agent

May 2026

- Built a **Kubernetes** platform that spins up agents on any **HuggingFace** model in one click, leveraging **NVIDIA Dynamo** for disaggregated prefill/decode serving, KV-cache-aware routing, and GPU autoscaling behind OpenAI-compatible endpoints

AdBlockIRL: AR ad-blocking proof-of-concept for Meta Quest

May 2025

- Trained **YOLOv8** on **Roboflow** to detect 100+ concurrent ads at 5ms latency, applying real-time mosaic censoring via **OpenCV**; won \$2000 at Bain Capital Ventures AI Hackathon

SlopGen: Automated short-form video generator

June 2024 - Sep 2024

- Built end-to-end pipeline using **yt-dlp**, **WhisperX**, and **llama3** via **ollama** achieving 92% segmentation accuracy (15x faster than manual), with custom TTS voices via **edge-tts**

EDUCATION

Rutgers University - New Brunswick

Bachelor of Science - Computer Science • Cum Laude

- Linux User Group (President), Road to Silicon Valley Program